Mohammad Javadian Farzaneh

PhD Student

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Google Scholar
Medium
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Education

PhD in Computer Science and Engineering — Sep 2024 - Ongoing
Sabanci University — *Istanbul, Turkey*

GPA : 3.83/4

Related Courses

- Computer Graphics (A-)
- Human Computer Interaction (A)
- Network Science (A)
- Data Mining (A)
- Machine Learning (B+)

Msc in Artificial Intelligence — Sep 2019 - Jan 2023
University of Isfahan — *Isfahan, Iran*

GPA : 17.42/20 (3.88/4)

Thesis : Implementation of a Blind navigation method in outdoors/indoors areas (Grade: 19.67/20 - Excellent)

Related Courses

- Neural networks (18.2/20)
- Pattern recognition (18.65/20)
- Digital image processing (17.5/20)
- Machine vision (16.7/20)

BSc in Computer Engineering — Sep 2014 - Feb 2019
Urmia University — *Urmia, Iran*

GPA : 15.58/20 (3.12/4)

Software Project : AI-Powered Backgammon Game for Android OS (Grade: 19.70/20)

Related Courses

- Artificial intelligence and Expert systems (16/20)
- Fund.of data mining and web search (17.5/20)

Research Interest

- Human-Computer Interaction, AR/VR/MR , Human-AI interaction
- Computer Vision, Human-Centered Design, Human-Robot interaction

Publications

Mohammad Javadian Farzaneh, Pelin Karadal, Baris Aygen, Kursat Cagiltay (2025). Evaluating usability gaps in passthrough mixed reality interfaces. International Journal of Human–Computer Interaction. Accepted September 2025. (IF: 4.9, Q1)

Mohammad Javadian Farzaneh, Selim Balcisoy (2026). Calibrated Trust and Resilience in Many-to-Many AI-Integrated XR Collaboration for Disaster Response. CHI'26: XR4CE workshop, April 14, 2026, Barcelona, Spain

Farzaneh, M.J. and Mohammadi, H.M., 2022. Implementation of a Blind navigation method in outdoors/indoors areas. arXiv preprint arXiv:2212.12185.

Projects

AI-Powered Backgammon Game for Android OS

Developed and implemented a dynamic backgammon AI with three difficulty levels: Easy (random moves), Medium (decision tree depth of one), and Hard (decision tree depth of four). Integrated strategic considerations, weighting moves based on board positions. Used an evaluation function to score and select optimal moves, enhancing strategic depth across difficulty levels.

Multi-Target Deep-Learning Sound Classification

Designed a sound classification system using MFCC-processed audio data and a CNN model. Utilized a dataset featuring sounds such as cat, dog, and traffic-related noises like fire trucks and ambulances.

Implementation of a Blind navigation method in outdoors/indoors areas

Designed a vision-based navigation system for the visually impaired using Visual SLAM and YOLO algorithm. Implemented obstacle detection, path deviation monitoring, and next-step prediction. Achieved a 30 fps frame rate with 8 cm positioning accuracy on a laptop camera. Successfully guided users within predefined 11-meter routes indoors and outdoors, issuing alerts for deviations exceeding 60 cm.

Face Swap

Developed a Python-based Face Swap application using Dlib library. Enabled effortless swapping of faces in images through facial landmark detection, convex hull extraction, and affine transformation.

Teaching Experience

Advanced Programming Sabanci University, Dr. Albert Levi in Spring 2025

Advanced Programming Sabanci University, Dr. Kamer Kaya in Fall 2024

Computer Graphics University of Isfahan, Dr. Mahvash in Spring 2023

Machine Vision University of Isfahan, Dr. Mahvash in Fall 2022

Microprocessor and Assembly language University of Isfahan, Dr. Mahvash in Spring 2022

Advanced Programming Urmia University, Dr. Shoja in Fall 2019

Work Experience

Unity Developer

Sep 2024

Tubitak *Istanbul,Turkey*

Developed a cross-platform, web-based game engine in Unity (C#) that enables teachers to create customized virtual educational environments, with mobile applications allowing students to seamlessly join and participate in sessions. Students using the mobile application can switch between a single-view mode and a side-by-side view compatible with Cardboard XR glasses.

Responsibilities:

- Researched and Developed web application and mobile application

- Integrated Microsoft technologies such as Azure and Playfab to manage application data
- Conducted weekly meetings with main supervisor and project faculty to maintain progress and assure project quality.
- Provided mentorship and technical guidance to Master's students within the project team.

Android Developer Apr 2020 – May 2022
Wachee *Tehran,Iran*

Working as an Android developer, I enhance and expand the capabilities of an app - a streaming VPN that empowers global users to enjoy Netflix on Android devices without encountering proxy errors.

Main Achievements:

- Developed proxy server using Netty framework on app to support HTTPS connections
- Mastered Android Materials and UI elements.
- Made usage of Jetpack libraries (e.g. AndroidX libraries).
- Used libraries such as RXJAVA or Kotlin coroutines to reach concurrency

Skills

English Proficiency TOEFL Overall: 105; R:29,L:28,S:22,W:26 June 2023

Programming Languages C#, C/C++, Java, Kotlin, Python, Matlab

Technology, Framework and Libraries Unity, WebXR, OpenCv, Opengl, LWJGL, Numpy, Pandas, Matplotlib, Sk-Learn, Pytorch

Build Tools Maven, Gradle

Project Managment Git, Trello

Typesetting Microsoft Office, Latex

Refrences

Dr. Selim Balcısoy: selim.balcisoy@sabanciunivedu office: +90 (216) 483 9569

Dr. Dilara Keküllüoğlu: dilara.kekulluoglu@sabanciuniv.edu office: +90 (216) 568 7042

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